AI-Enabled Coconut Land Utilization, WCLWD Severity Assessment, Germplasm Identification and Community Intelligence Advisory System

Abstract*—Coconut gene bank conservation and plantation management involve four largely disconnected information needs: identifying germplasm from limited field images, estimating plantation land utilization from aerial imagery, assessing Weligama Coconut Leaf Wilt Disease (WCLWD) severity, and providing accessible disease-management guidance. This paper presents an integrated web platform (React, Node.js/Express, Flask, MySQL) composed of four independently developed and evaluated AI modules accessible through a common interface. The germplasm-identification module applies an EfficientNet-B0 embedding network within a Prototypical Network under a 3-way, 8-shot, 5-query few-shot protocol to a primary field dataset of 428 images collected at the Coconut Research Institute (CRI), Bandirippuwa, reaching 96.53% final episodic evaluation accuracy. The land-utilization module fine-tunes a YOLO11n detector on a public aerial Palm/Tree dataset (mAP@50 = 83.5%) and combines detections with Excess-Green vegetation masking and a Euclidean-distance-transform-based greedy algorithm to estimate image-space occupied area and approximate additional planting positions. The WCLWD severity module fine-tunes EfficientNet-B0 for four-class leaf classification (98.79% test accuracy) and couples Grad-CAM-guided weakly supervised pseudo-segmentation with an ABCD-inspired weighted Disease Severity Index for computational severity categorization. The disease-identification and advisory module classifies four coconut leaf conditions (98.89% test accuracy), screens unfamiliar inputs using a 256-dimensional Mahalanobis out-of-distribution check, and generates bilingual (English/Sinhala) management guidance through a retrieval-augmented generation pipeline over a curated 16-document knowledge base. The four modules retain independent datasets, pipelines, and evaluation designs, no combined accuracy is computed. Results indicate that the individual components are promising, while broader field, expert, and out-of-distribution validation remain necessary before operational deployment.

***Index Terms*—coconut gene bank, germplasm identification, few-shot learning, land-utilization analysis, YOLO, Weligama coconut leaf wilt disease, weakly supervised segmentation, out-of-distribution detection, retrieval-augmented generation**

I. Introduction

Coconut (*Cocos nucifera* L.) is a plantation crop of major economic importance in tropical countries such as Sri Lanka, where germplasm conservation, plantation land management and crop-health monitoring are traditionally performed through manual visual inspection and specialist judgement. Germplasm identification requires trained personnel, plantation-utilization assessment requires physical land inspection, Weligama Coconut Leaf Wilt Disease (WCLWD) is frequently confused with nutrient-deficiency symptoms and other physiological disorders [1], [2], and disease-management knowledge is fragmented across extension officers, printed material and informal sources. These four tasks are manual, time-consuming, and difficult to scale.

Advances in deep learning, few-shot image classification, object detection, explainable AI, weakly supervised segmentation, out-of-distribution (OOD) detection and retrieval-augmented generation (RAG) offer complementary tools for each of these tasks. Rather than a single end-to-end model, this project investigates four distinct AI-based decision-support components unified at the application level. The project-level gap addressed is the limited integration of germplasm identification, land-utilization analysis, WCLWD severity assessment, and disease advisory support within one coconut-focused platform, this is not a claim that no prior AI-based coconut research exists.

This paper introduces four complementary modules: A few-shot image classifier for coconut germplasm identification, an aerial-image land-utilization and approximate planting-capacity estimator, A WCLWD classification, explainability, and computational severity-assessment pipeline, and an OOD-aware disease classifier coupled with a bilingual RAG advisory chatbot. The overall research aim is to design, implement, and evaluate these four modules and integrate them through a common React/Node.js/Express/Flask/MySQL web platform. The principal contributions are: few-shot approach applied to a locally collected germplasm dataset, transformation of aerial Palm/Tree detections into image-space utilization and capacity estimates,integration of classification, explainability, weakly

supervised pseudo-segmentation and computational severity scoring for WCLWD, and integration of disease classification, explicit OOD screening and curated bilingual RAG advisory generation. This work does not claim to introduce EfficientNet, Prototypical Networks, Grad-CAM, YOLO, Excess Green indexing, GrabCut, Mahalanobis out-of-distribution (OOD) detection, or retrieval-augmented generation (RAG). Instead, its primary contribution is the project-level integration of four independently developed decision-support modules: a few-shot classifier with Grad-CAM visualization for identifying coconut germplasm using a field-collected dataset, an image-space transformation of aerial Palm/Tree detections into land-utilization and planting-capacity information, an integrated WCLWD classification, Grad-CAM visualization, pseudo-segmentation, and severity-scoring pipeline, and an OOD-screened disease classifier coupled with a curated-knowledge bilingual RAG advisory system. A further contribution is the unified application-level access provided to all four modules through a single platform. The remainder of the paper is organized as follows: Section II reviews the literature, Section III describes the methodology of the four modules, Section IV presents the results, Section V discusses the findings, limitations and future work, and Section VI concludes the paper.

II. Literature Review

A. Germplasm Identification and Few-Shot Classification

Convolutional networks such as EfficientNet [3], pretrained on ImageNet [4], are widely used as transfer-learning backbones for agricultural image classification. When labelled data are scarce, few-shot approaches such as Prototypical Networks[5] classify a query by its distance to class prototypes computed from a small support set, which is well suited to a primary, field-collected germplasm dataset of a few hundred images per class.

B. Aerial Palm/Tree Detection and Land-Utilization Assessment

Object detectors of the YOLO family [6] are commonly applied to UAV and aerial imagery for palm-tree detection and counting [7]. Reported detector performance varies with imaging conditions and IoU thresholds, classical image-processing cues such as the Excess Green vegetation index and the Euclidean distance transform are used to complement detector output for coarse occupancy and free-space estimation when georeferenced GIS layers are unavailable.

C. WCLWD Classification, Explainability and Severity Assessment

WCLWD is a phytoplasma-associated disease confined mainly to southern Sri Lanka [1] whose symptoms are difficult to distinguish visually from nutrient deficiency and other disorders [2]. Grad-CAM [8] localizes discriminative image regions and has been used as a basis for weakly supervised segmentation when manual pixel annotations are unavailable, typically refined using Otsu thresholding [9] and GrabCut [10]. ABCD-inspired severity indices, following dermatological precedent, combine shape and area descriptors into a single index.

D. Out-of-Distribution-Aware Agricultural Classification

Mahalanobis-distance-based OOD detection [11] models class-conditional feature distributions and flags inputs that fall outside the learned distribution, which is important for field deployment where an image may not belong to any trained class.

E. Knowledge-Grounded and Bilingual Agricultural Advisory

Retrieval-augmented generation [12] grounds large-language-model responses in retrieved document chunks rather than parametric memory alone, reducing (but not eliminating) hallucination. Dense retrieval commonly relies on general-purpose text embedding models such as BGE [13]. Prior WCLWD-focused work has combined CNNs and detectors for disease and pest identification and severity scoring [2]. Domain literature such as reviews of the coconut leaf rot disease complex [14] and field studies of farmers' adoption of recommended pest-management technologies
[15] illustrate the kind of agricultural-extension knowledge such advisory systems can draw on.

F. Synthesis

Individually, few-shot classification, aerial object detection, explainable disease-severity pipelines and RAG-based advisory systems are established directions. The specific gap this project addresses is their combined, application-level availability for coconut germplasm and crop-health management within one platform, rather than the invention of any individual algorithm.

III. Methodology

The four modules are independently trained AI components – they are not chained into a single pipeline, and each accepts its own input and produces its own output through the shared web platform. Fig. 1 summarizes, for each module in order, the input it requires, its role, and the output it returns, before the module-by-module technical description that follows.

A. Few-Shot Image Classification for Coconut Germplasm Identification

This model recognizes coconut germplasm as Dwarf, King Coconut, or Tall based on a primary dataset comprising 428 field images captured at Coconut Research Institute (CRI), Bandirippuwa, Sri Lanka. The process starts with converting the input image into RGB format, resizing it to 224×224 pixels, and normalizing it using ImageNet parameters. Training data images undergo data augmentation techniques, including horizontal flip, rotation, and color adjustment.

The image undergoes feature extraction and projection into a 512-dimensional embedding space using the ImageNet-pretrained EfficientNet-B0 backbone network. This process is conducted in an episodic fashion using the Prototypical Network method with a 3-way, 8-shot, and 5-query setting.

TABLE I
GERMPLASM IDENTIFICATION DATASET AND FEW-SHOT TRAINING CONFIGURATION

Item	Value
Images / classes	428 (Dwarf 118, King Coconut 67, Tall 243)
Train / val. pool	341 / 87 images
Episode config.	3-way, 8-shot, 5-query
Training	40 epochs, 40 train / 15 val. episodes per epoch
Final eval.	50 episodes (750 query instances)

TABLE II
LAND-UTILIZATION DETECTOR PERFORMANCE AND DEMONSTRATION-IMAGE STATISTICS

Item	Value
Detector/data	YOLO11n, 279 images, 10,398 boxes (Palm, Tree)
Precision/Recall	78.2% / 82.4%
mAP@50/50–95	83.5% / 48.9%
Detections (demo)	86 Palm/Tree objects
Circular util./avail.	32.29% / 67.71%
Refined util./avail.	23.81% / 76.19%
Candidates/capacity	172 candidates / ≈55 positions

For every episode, eight support images per class are used to compute the class prototype by averaging embeddings, and each query image is mapped against three class prototypes using cosine similarity with learnable scales. The output of cosine similarity is converted to probability by applying the softmax function, and this model is optimized via cross-entropy loss.

In the testing phase, the embedding of a newly uploaded coconut image is mapped against the computed class prototypes. The class having the highest probability is returned with its confidence level and probability scores per class. Grad-CAM visualization technique is then performed to obtain the image regions responsible for making such a prediction. The model is deployed with the Flask backend for image recognition through the web interface.

B. Aerial Land-Utilization and Approximate Capacity Analysis

This module estimates plantation occupancy and approximate additional planting capacity from a single aerial image. A YOLO11n detector was fine-tuned on the public Kaggle Aerial Images of Palm Trees dataset (Palm/Tree classes, 279 annotated images, 10,398 boxes, 237 training / 42 validation images, 50 epochs, 640×640 input, batch size 8). For a given input image, detected boxes and their centers seed an initial Shapely circular-buffer occupancy estimate, which is refined with a dynamic Excess Green (ExG = $2G-R-B$) vegetation mask combined with elliptical constraints and morphological cleaning to obtain a refined utilized/raw-available area split. A Euclidean Distance Transform locates local-maximum free-space candidates, and a greedy algorithm applies an image-space spacing proxy to approximate additional planting positions. All quantities are in pixel/image space no GSD, georeferencing, or metric GIS layer was incorporated in the current implementation.

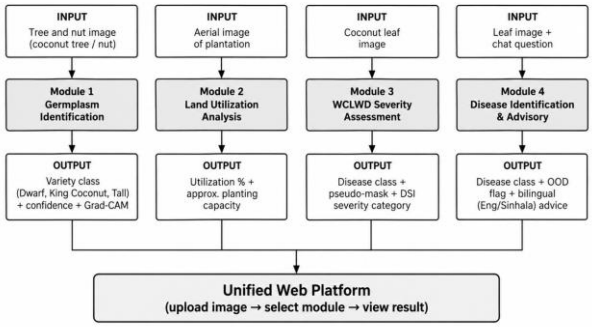


Fig. 1. System input/output overview: for each of the four modules, the image (and, for Module 4, chat query) submitted by the user, and the result returned through the shared web platform.

TABLE III
WCLWD CLASSIFICATION ACCURACY AND SEVERITY-INDEX CONFIGURATION

Item	Value
Dataset/test	3,354 images / 497 test images
Test accuracy	98.79%
Weighted P/R/F1	98.80% / 98.79% / 98.77%
Severity index	Weighted DSI, Eq. (1), 4 severity categories

C. WCLWD Classification, Explainability and Computational Severity

This module transforms a coconut leaf image into a four-class prediction (Healthy Leaves, WCLWD Drying of Leaflets, WCLWD Flaccidity, WCLWD Yellowing), a Grad-CAM explanation, a weakly supervised disease-region pseudo-mask, and a computational severity score. The Coconut Diseases and Pest Infestations dataset (3,354 deduplicated images via perceptual hashing and group-based splitting, training-only oversampling of the minority Healthy class) was preprocessed with denoising, LAB/CLAHE contrast enhancement and ImageNet normalization, then used to fine-tune EfficientNet-B0. A late-layer Grad-CAM map seeds image-specific Otsu thresholding and GrabCut refinement, followed by morphological cleaning, to produce a pseudo-segmentation mask – not manually annotated ground truth. ABCD-inspired features (border irregularity A, solidity-related term B, distribution C, diameter/extent D) and an affected-area fraction are combined, after 5th/95th-percentile calibration, into a weighted Disease Severity Index:

$$\mathrm{DSI} = 0.20A + 0.20B + 0.20C + 0.15D + 0.25 \cdot A_{\mathrm{affected}} \quad (1)$$

where A, B, C and D are the calibrated ABCD-inspired descriptors and Affected is the calibrated affected-leaf-area fraction. DSI is mapped to Mild ($\leq$25), Moderate (25–50], Severe (50–75] or Critical (>75) categories, a Healthy prediction is assigned DSI = 0. These thresholds are computational and have not been clinically/agronomically validated.

TABLE IV
DISEASE CLASSIFIER, OOD-DETECTION AND RAG ADVISORY CONFIGURATION

Item	Value
Test acc./loss	98.89% / 0.3841
OOD threshold	594.53 (256-D Mahalanobis, 99th pct.)
Known-ID flagged	0.97%
RAG chunks/dim.	1,606 chunks, 1,024-D (BGE-large-en-v1.5)
Retrieval/LLM	Top-5, Pinecone, GPT-4o-mini via Open-Router

D. OOD-Aware Disease Classification and Bilingual RAG Advisory

This module classifies coconut leaf images into CCI_Caterpillars, Gray Leaf Spot, Healthy_Leaves and Leaf Rot, screens predictions for unfamiliar inputs, and provides bilingual (English/Sinhala) management guidance for Coconut Caterpillar Infestation, Coconut Leaf Rot and Coconut Gray Leaf Spot. An EfficientNet-B0 classifier produces a custom 256-dimensional representation from an original 4,807-image dataset (3,364/720/723 train/validation/test before balancing, balanced to 8,000 training images at 2,000 per class). A Mahalanobis-distance OOD stage compares class-conditional means and a shared precision matrix against a 99th-percentile threshold. The advisory knowledge base consists of 16 curated coconut-disease PDF documents – including plant-pathology literature on the coconut leaf rot disease complex and a Coconut Research Institute (CRI) field study on Sri Lankan farmers' adoption of recommended pest-management technologies, alongside other curated agricultural-extension and disease-management sources – split into 1,606 chunks of 900 characters with 150-character overlap, embedded with BAAI/bge-large-en-v1.5 (1,024-dimensional), indexed in Pinecone, and retrieved (top-5) as context for openai/gpt-4o-mini via OpenRouter, with English/Sinhala language detection, translation and conversational memory. This final architecture replaced an earlier crowdsourcing concept because of reliability concerns, crowdsourcing is project history, not the deployed advisory mechanism.

IV. RESULTS

The four modules were implemented and deployed as a single-page web application, "Coconut Diagnostics Suite," with Variety Classifier (Module 1, germplasm identification), Canopy & Land Survey (Module 2, land utilization), Leaf Wilt Diagnostic (Module 3, WCLWD severity) and Treatment Advisor (Module 4) as its four independent workflows, accessed through a unified React dashboard backed by a Node.js/Express server, a Flask inference layer, and MySQL storage, consistent with the input/output overview in Fig. 1. Each module operates independently and returns its own result through this shared interface. The following subsections report each module's held-out performance together with a qualitative demonstration of its output.

A. Germplasm Identification Results

The Prototypical Network reached a best episodic validation accuracy of 98.67% at epoch 27 and a final episodic evaluation accuracy of 96.53% (724/750 correct query-instance predictions across 50 episodes), with overall precision, recall and F1-score of 0.97. Fig. 2(a) shows the corresponding confusion matrix, confirming that the 26 misclassifications are concentrated between Dwarf and Tall rather than involving King Coconut. Fig. 2(b) plots training query accuracy across the 40 training epochs, rising from about 0.76 to near 1.00 within the first 10 epochs. A qualitative single-image example was classified as King Coconut with 99.97% confidence – a per-image confidence value, not the overall accuracy – and the Grad-CAM visualization in Fig. 2(c) highlights the fruit region used for the prediction, without expert validation.

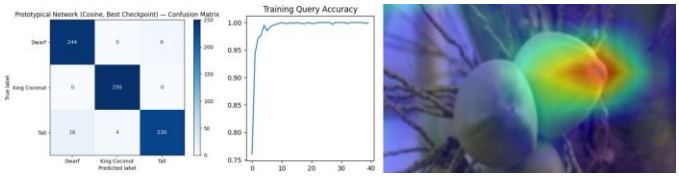


Fig. 2. Germplasm identification (Module 1): (a) confusion matrix on 750 episodic query instances, (b) training query accuracy across 40 episodes, and (c) Grad-CAM visualization for a single classification example.

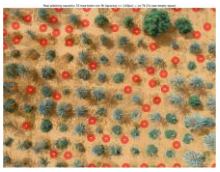


Fig. 3. Land-utilization (Module 2) demonstration: detected palms and 55 numbered candidate planting positions overlaid on the raw available (76.2%) space of the input aerial image.

B. Land-Utilization Results

Detector validation gave mAP@50 = 83.5% and mAP@50–95 = 48.9% (precision 78.2%, recall 82.4%), this is a detection metric, not classification accuracy. On the single demonstration image, 86 Palm/Tree detections yielded 32.29% circular / 23.81% refined vegetation-mask utilization, 76.19% raw available (non-plantable-guaranteed) area, 172 free-space candidates and approximately 55 additional positions – image-space estimates, not field-validated planting recommendations. Fig. 3 shows this demonstration image, with detected palms and the 55 numbered candidate planting positions (red markers) overlaid on the raw available space.

C. WCLWD Assessment Results

On 497 held-out test images, the classifier reached 98.79% accuracy (weighted precision 98.80%, recall 98.79%, F1 98.77%). Per-class recall was lower for the minority Healthy Leaves class (0.84) than for the three WCLWD classes (0.99–1.00), reflecting class imbalance, as shown in the confusion matrix of Fig. 4(a), Fig. 4(b) plots the corresponding training/validation accuracy curve across 20 epochs. Grad-CAM-to-Otsu-to-GrabCut pseudo-masks and DSI values were produced per image, no IoU/Dice against manual ground truth or expert-validated severity accuracy is reported. Fig. 5 illustrates this pipeline on a single WCLWD_Yellowing example: the

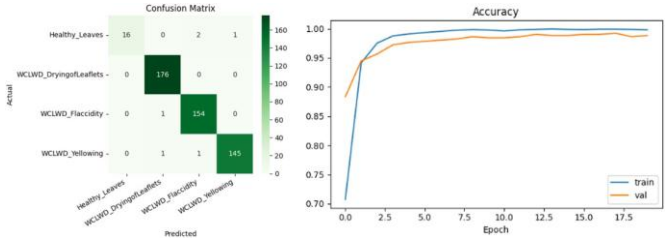


Fig. 4. WCLWD assessment (Module 3) classifier performance: (a) confusion matrix on 497 held-out test images and (b) training and validation accuracy across 20 epochs.

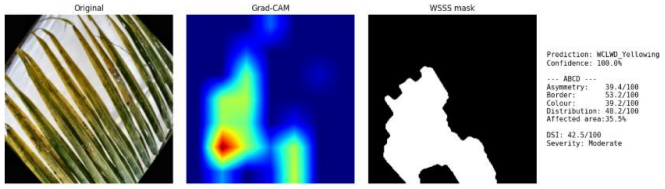


Fig. 5. WCLWD severity pipeline for a *WCLWD Yellowing* example: (a) original image, (b) Grad-CAM heatmap, (c) WSSS binary mask, and (d) resulting ABCD descriptors with a DSI of 42.5, indicating moderate severity.

Grad-CAM heatmap, the resulting WSSS binary mask, and the computed ABCD descriptors (Asymmetry 39.4, Border 53.2, Colour 39.2, Distribution 48.2, affected area 35.5%) combine into a DSI of 42.5/100, categorized as Moderate severity per Eq. (1).

D. Disease Identification, OOD and RAG Results

The four-class disease classifier reached 98.89% test accuracy (loss 0.3841) on 723 held-out images. Fig. 6(a) gives the confusion matrix, Fig. 6(b) the training/validation accuracy curve (with the fine-tuning phase marked), and Fig. 6(c) the distribution of Mahalanobis distances on in-distribution validation/test data relative to the 99th-percentile OOD threshold (594.53). With this threshold, 0.97% of known in-distribution test samples were flagged as OOD, this is a false-flag rate, not an OOD accuracy, and no external OOD dataset was tested. In a single qualitative example, an input leaf was classified as CCI_Caterpillars with 93.3% confidence and a Mahalanobis distance of 310.79, well below the OOD threshold. The bilingual RAG pipeline (16 PDFs, 1,606 chunks, top-5 retrieval, GPT-4o-mini) produced functional English/Sinhala responses in manual demonstrations – for example, a Sinhala-language response for coconut caterpillar infestation recommending chemical control (Folidol Dust, Contaf 5EC, Indofil M-45), biological control (Trichogramma release), manual removal and cultural practices – but these are not a formal chatbot evaluation and do not establish factual guarantees.

V. Discussion

A. Discussion of Results

Module 1 shows that few-shot learning can support germplasm identification from a modest dataset, but the 750 evaluated instances are episodic appearances of the same 428 images rather than independent external samples. Module 2 illustrates the difference between circular occupancy,

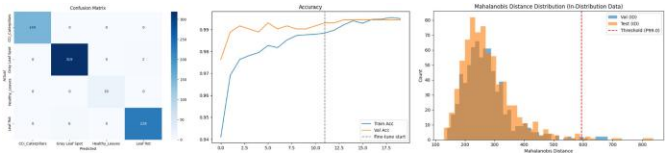


Fig. 6. Disease and OOD classifier performance (Module 4): (a) confusion matrix on 723 held-out test images, (b) training and validation accuracy with the fine-tuning phase marked, and (c) Mahalanobis-distance distribution relative to the 99th-percentile OOD threshold.

TABLE V
Consolidated Accuracy and Key Results Across the Four Modules

Module	Method	Key Result	Limitation
1. Germplasm	EfficientNet-B0+ProtoNet, few-shot	96.53% episodic acc.	Single site, no indep. test
2.Land Util.	YOLO11n + ExG + dist. transform	mAP@50 = 83.5%	Pixel-space, no GSD/GIS
3. WCLWD	EfficientNet-B0 + WSSS + DSI	98.79% test acc.	Mask/DSI un-validated
4.Disease+ RAG	EfficientNet-B0 OOD + RAG/LLM	98.89% test acc.	No ext. OOD, RAG unguar-anteed

vegetation-mask-refined utilization, candidate free-space pockets and spacing-constrained capacity – four related but distinct image-space quantities, none of which yet constitute field-validated planting guidance. Module 3 combines strong overall classification performance with a lower Healthy-class recall and an unvalidated pseudo-segmentation/DSI pathway, underscoring that explainability outputs are diagnostic aids rather than ground truth. Module 4 pairs high in-domain accuracy with only a preliminary, in-distribution OOD check and an advisory layer whose retrieval grounding reduces but does not eliminate the risk of incorrect LLM output. Because the four modules use unrelated datasets, modalities and metrics, their results are not directly comparable and no combined system-level score is computed. Considered together, the four modules demonstrate that heterogeneous AI techniques can be integrated behind one platform to cover complementary coconut-management needs, while each module's evidence base must be strengthened before field deployment.

B. Limitations

Dataset/sampling: Module 1's dataset is single-site and class-imbalanced, Module 2 uses a public aerial dataset not collected at a coconut gene bank, Modules 3–4 datasets are drawn from limited sources. Evaluation: none of the four modules has an independent, externally sourced test set, Module 1's instances are episodic re-appearances rather than unique external images. Explainability: Grad-CAM/WSSS outputs in Modules 1 and 3 are not expert-validated. Spatial: Module 2 operates in pixel space without GSD or agronomic layers. OOD: Module 4's check used only known in-distribution samples. RAG/translation: Module 4's responses are not guar-anteed factually correct, and Sinhala translation quality is unevaluated. The platform has not undergone field testing or operational deployment.

C. Future Work

Module 1: multi-location data collection and external testing. Module 2: georeferenced imagery, GSD, GIS coordinates and agronomic layers, with comparison to dedicated segmentation models. Module 3: expert validation of Grad-CAM/pseudo-mask output and of DSI against agronomic severity. Module 4: a dedicated external OOD dataset and formalretrieval-quality/hallucination-rate/translation evaluation. System level: multi-module field trials, usability studies with CRI stakeholders, and production deployment.

VI. Conclusion

This paper presented an integrated platform combining four independently developed AI modules for coconut germplasm identification, land-utilization analysis, WCLWD severity assessment, and disease advisory support, achieving 96.53% episodic evaluation accuracy, mAP@50 = 83.5%, 98.79% and 98.89% held-out classification accuracy, respectively. The project-level contribution lies in integrating these complementary modules behind one common platform rather than in any single algorithm. Broader field, expert, OOD and deployment validation remain necessary before the system can be relied upon operationally.

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